

World's smallest GPS/GLONASS receiver

ST-Ericsson claims to have come up with the world's smallest receiver that will be able to see both GPS and GLONASS positioning satellites, enabling mobile devices to deliver much faster and reliable location-based services.

Compared to GPS-only solutions, the combination of GLONASS and GPS technologies improves positioning accuracy in urban areas, where tall buildings can obscure satellite signals, and also reduces the time taken to initially determine the location of a device.

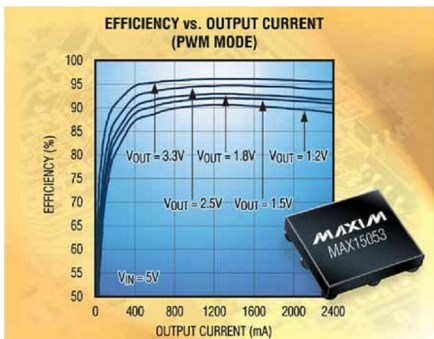
Requiring just two external components, the CG1950 is said to be the first GPS/GLONASS receiver to be built in 45nm silicon, enabling manufacturers to produce sleek, low-cost devices capable of supporting highly accurate navigation, mobile social networking, augmented reality and other location-based services. The low-power CG1950 is designed to be integrated into mobile phones, camcorders, cameras and other mobile devices.

Smallest 2A synchronous buck

Maxim Integrated Products has introduced a current-mode, synchronous DC-DC converter in a tiny $1.65 \times 1.65 \text{ mm}^2$ wafer-level package. This miniature step-down regulator integrates MOSFETs to simplify design, minimise electromagnetic interference and save board space while achieving conversion efficiency of up to 96 percent at full load (2A).

The MAX15053 operates from a fixed 1MHz switching frequency that allows for small external passive components and an all-ceramic-capacitor design, further minimising total solution size. A peak-current-mode architecture ensures excellent load regulation while simplifying compensation.

Offering both fixed pulse-width modulation and skip-mode operation, the MAX15053 is suitable for applications ranging from telecom, network-



Industry's smallest 2A synchronous buck integrates MOSFETs from Maxim

ing and base-station equipment to portable devices such as notebooks.

Industry's first open mobile core

Juniper Networks claims to have announced the industry's first open mobile core for 2G/3G and LTE networks that might enable mobile operators to profit from the rapid increase in mobile data traffic from smartphones and other mobile devices.

Called MobileNext, it is a suite of software that runs on Juniper Networks' 3D universal edge router and Junos software platform. With scalability of mobile sessions, set-up rates and forwarding capacity, it enables high-definition voice and video delivery with handoff between 3G and LTE for non-interrupted user experience while integrating a rich portfolio of in-line IP services. These in-line services, including deep packet inspection, traffic direct for internet offload, carrier-grade NAT, firewall, video optimisation, multiprotocol label switching and application load balancing, provide deployment flexibility and dramatically reduce the total cost of

ownership to mobile operators.

Test solution for PCI Express 3.0 receiver characterisation

Agilent Technologies claims to have developed the industry's first test set for PCI Express 3.0 receiver characterisation. The test solution provides complete and accurate receiver tolerance test results while minimising R&D effort. Using the Agilent test set, design and test engineers in the semiconductor and computer industry can accurately characterise and verify standard compliance of receiver ports in ASICs and chipsets.

The PCIe 3.0 receiver test solution includes high-performance serial BERT, de-emphasis signal converter, pulse function arbitrary noise generator, high-performance oscilloscope, new accessories and new test automation software. Benefits include accurate and repeatable receiver test results enabled by new stress calibration software, adjustable pre- and post-cursor de-emphasis, J-BERT's built-in PCIe 3.0-compliant jitter and sinusoidal interference sources, new PJ sweep functionality and new PCIe 3.0-compliant calibration channels.

Check efytimes.com for more news, daily